a Scientific input & delivery DAG

Research intent

Molecules · reaction conditions

Competing pathways & methods

Natural-language hypothesis formation

Delivery DAG



Species nodes · reaction-step edges

Default: ts_opt · frequency · short_irc

Step query

b Geometry & PES

Atom mapping

Atom-conserving candidates

Reaction-center atom pairs

Steric traversal

Distance-weighted local bulk

Substituent contributions, ρ^{d}

Rigid alignment

Kabsch endpoint alignment

Consistent endpoint frames

Path preparation

NEB–IDPP · light IDPP

Candidate paths & TS seeds

Job dispatch

Evidence binding

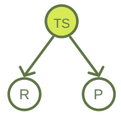
Guided diagnostics

Cloudra

Cloudra Agent System

d Mechanistic validation & refinement

Identity-resolved pathways



Same TS → ±IRC → Opt/Freq

Atom mapping · species identity

$n_{imag}(TS) = 1 \cdot n_{imag}(R/P) = 0$

Reaction-centered modes

Pair projections · atomic motion

Reaction-center contributions

Evidence-guided revision

Locate conflicting evidence

Revise affected calculations

Native outputs

c Computational infrastructure

Scheduler

Per-node quotas Pinned hosts

Independent of the web API

ComputeDB ledger

Input hashes Deduplication

Durable records · recovery

Engine execution

ORCA · Gaussian · xTB

Native logs · file locks

Geometries · energies · frequencies